Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Management of Persistent Respiratory Symptoms in Pre-School Children

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1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 Recurrent wheeze and cough symptoms affect a third of children aged <5 years (1) which present a significant burden of morbidity to healthcare services during the pre-school years. Triggers include viral infections, exposure to tobacco smoke and environmental pollutants. Bronchiolitis can cause airways hyper-reactivity and respiratory symptoms for many years (2).

1.3 Whilst the majority of these children do not have an increased risk of asthma or allergy in later life (3) there *is* a progression from childhood cough and wheeze symptoms to adult asthma in *some* children (4).

1.4 **Controversy and confusion exist over which treatments are effective in managing recurrent wheeze and cough in pre-school children.**

2. MANAGEMENT IN PRIMARY CARE

2.1 The majority of -re-school children with respiratory symptoms lasting >4 weeks in duration can be managed in primary care by their General Practitioner as long as there are no worrying features (**Table 1**) suggestive of other diagnoses (**Table 2**).

2.2 There should be a low threshold for referral into hospital if there are worrying symptoms/signs, uncertainty about the underlying diagnosis or if there are medical/parental concerns.

2.3 Management of the otherwise well child with a persistent dry cough will include reassurance and a “watch and wait” approach.

2.4 Chronic respiratory symptoms can cause significant distress to the whole family. It is therefore important to explore the reasons for parental anxiety and to reassure parents that a watch and wait approach is probably all that is needed.

Table 1: Worrying features suggestive of underlying lung disease in children with chronic respiratory symptoms:

Neonatal onset of respiratory symptoms
Recurrent episodic / moist cough lasting in excess of 4-6 weeks
Onset after choking episode
Occurs during / after feeding
Auscultatory findings i.e clear with coughing
Cardiac abnormalities
Chest wall deformity
Immune deficiency
Feeding difficulties
Haemoptysis
Neurodevelopmental abnormality
Faltering growth / general ill health
Finger clubbing
Recurrent lower respiratory tract infections – *See " Management of Community Acquired Pneumonia" guideline.*

Table 2: Differential diagnosis of recurrent wheeze and cough in pre-school children:

Cigarette smoke exposure / air pollution
Chronic lung disease of prematurity
Aspiration (dysfunctional swallow, trache-oesophageal fistula / laryngeal cleft etc)
Cystic fibrosis
Tuberculosis
Inhaled foreign body
Tracheobronchomalacia / congenital abnormality of lung, airway or heart
Immune deficiency
Primary ciliary dyskinesia
Psychogenic

3. MANAGEMENT IN HOSPITAL

3.1 Investigations *following referral* to hospital

- Oxygen saturations – admit for assessment and onward management if ≤ 92% in air, taking <50% of usual feeds for ≥24 hours, signs of fatigue or CO2 retention (sweaty, irritable) or a history of apnoeas.
- Chest x-ray (proceed to HRCT chest following discussion with either Dr Emily Bell or Dr Beth Enderby if abnormalities other than those of an acute infection are detected).
- Cough swab (sputum) for MCS and viral studies (consider per nasal swab).
- *Where there are worrying features suggestive of a chronic lung disease process, the child should be discussed with either Dr Emily Bell or Dr Beth Enderby and the following investigations should be considered:*
    - Sweat – test

- Immunoglobulins: Ig G and Ig G subclasses, Ig M, IgA and Ig E
- Functional antibody titres to *Pneumococcus* and *Haemophilus influenza B a – 1 anti-trypsin titres.*
- Skin prick testing to aeroallergens (grass pollen, tree pollen, house dust mite, moulds, cat and dog) and other possible triggers identified by parents.
- Cillial brushings – arrange via Dr Beth Enderby or Dr Emily Bell.

3.2 Treatment

- Treatment is largely supportive. Parents should be advised about signs of increasing respiratory effort, fatigue and dehydration. They should adopt a low threshold for presentation to acute paediatric services if they are at all concerned about their child.
- There are no disease-modifying therapies for wheezing and cough in pre-school children. Treatment in this age group is therefore aimed at relief of symptoms to allow the child and his/her family to lead as normal a life as possible with the use of minimal medication and as little disruption to family life as possible.
- Any trial of therapy should be an appropriate dose for a sufficient duration of time (8-10 weeks as a minimum).

Antibiotics:

- Antibiotics should be reserved for situations where a bacterial infection is strongly suspected – see *"Management of community acquired pneumonia in children" guideline.*

Bronchodilators:

- A Cochrane systematic review (5) found that, although inhaled salbutamol was no better than placebo in reducing the frequency of non-specific cough in children, a sub-group of children might benefit from its use.
- Short-acting ß-2 agonists are the treatment of choice for acute symptoms of wheeze. A trial of salbutamol (6 x 100mcg puffs via MDI and spacer) should be given and only repeated if there is a reported benefit in respiratory effort or wheeze/cough symptoms 10-20 minutes following use. If found to be of benefit, the standard salbutamol dose is 2x100mcg puffs via MDI and spacer, used twice daily and up to a maximum of every four hours.
- **Any child requiring salbutamol more than every four hours should be assessed by medical staff.**
- There is insufficient evidence to support the use of **anticholinergics** (6) (e.g. ipratropium bromide) and adverse events such as a paradoxical bronchospasm and tachycardia are well-documented.

Inhaled corticosteroids:

- The clinical impact of inhaled corticosteroids (ICS) on chronic cough/wheeze symptoms in pre-school children is questionable (7).
- Given that ICS have been shown to be effective in the management of the majority of childhood asthmatics, the use of 400µg/ day equivalent of budesonide (i.e. low-dose) can be trialled for a 6-8 week period.
- Where cough symptoms resolve during this trial, the child may not necessarily have asthma and should be re-evaluated off of ICS.
- **Cough unresponsive to ICS should not be treated with increasing doses of ICS.**

Oral corticosteroids

- There are no randomised controlled trials looking at the use of oral steroids for recurrent chest symptoms in children <5 years.
- These can be used during exacerbations if the child has either had a previously good response to these or if they have never been given them previously.
- Where oral steroids are used, an ICS should be trialled for a 6-8 week period following discharge from hospital.

Leukotriene-receptor antagonists (LTRAs):

- Leukotrienes are pro-inflammatory mediators that are produced by a range of cells, including eosinophils, mast cells and alveolar macrophages, in response to stimuli in the airway. The cysteinyl leukotrienes (C4, D4 and E4) and induce bronchoconstriction, eosinophil chemotaxis, tissue oedema and production of secretions (8). LTRAs (montelukast) block cysteinyl leukotriene receptors.
- When given regularly, LTRAs are of proven benefit in pre-school children with persistent wheeze (9).
- A Cochrane review is in progress to evaluate the effect of regular versus intermittent LRTAs in the management of episodic viral induced wheeze in children aged <6 years (10) as evidence suggests that LTRAs given at the start of an exacerbation may be more effective than regular use (11).

Nebulised hypertonic saline

- In a small single study, nebulised hypertonic saline has been shown to shorten length of hospital stay and hospital admission rate in -reschool children presenting with acute wheeze (12). The benefits of its regular use as a mucolytic and anti-inflammatory agent in children with recurrent viral induced wheeze is unknown but it is generally well-tolerated and its use has produced some anecdotal benefit in children with very resistant symptoms (3% mucoclear or 7% nebusal both given 4ml twice daily).

- Referral to the paediatric respiratory team *should* be made if this treatment is being considered to a? ensure parent education and an appropriate level of onward support/supervision and b? so that a trial dose can be given to assess if this treatment is safely tolerated.

4. FOLLOW-UP

4.1 Children commenced on any of these treatment strategies should be reviewed at 6-8 week intervals to assess progress. Ideally, this period of treatment should include an upper respiratory tract infection to assess effectiveness of treatment.

4.2 Where these treatments are felt to produce no improvement in symptom control or are poorly tolerated, then they should be stopped and another treatment trialled for a further 6-8 week period of time.

4.3 Parents should be advised to return to their GP if there are increasing concerns and a low threshold should be maintained for onward referral to the paediatric respiratory team.

4.4 Cessation of parental smoking has been shown to be a successful therapy for respiratory symptoms in children.

5. REFERENCES

1. Bisgaard H, Szeffer S. Prevalance of asthma-like symptoms in young children. Pediatr Pulmonol 2007;42:723-8.
2. Fjaerli HO, Farstad T, Nakstad B. Acute bronchiolitis in infancy as risk factor for wheezing and reduced pulmonary function by seven years in Akershaus County, Norway. BMC Pediat 2005. 18 (5):31
3. Martinez FD, Wright AL, Taussig LM et al. Asthma and wheezing in the first six years of life. The Group Health Medical Associates. N. Engl J Med 1995;332:133-8
4. Sears MR, Greene JM, Willan AR et al. A longitudinal population-based cohort study of childhood asthma followed to adulthood. N. Engl. J Med 2003;349:1414-22.
5. Tomerak AAT, Vyas H,Lakhanpaul M et al. Inhaled beta 2- agonists for non-specific cough in children. Cochrane Database Syst. Rev 2005;3:CD005373
6. Chang AB, MckeanM, Morris P. Inhaled anticholinergics for prolonged non-specific cough in children. Cochrane Database Syst. Rev 2003;4:CD004358
7. Tomerak AA, Mc Glashan JJM, Lakhanpaul M et al. Inhaled corticosteroids for non-specific cough chronic cough in children. Cochrane Database Syst. Rev. 2005;4:CD004231.Pub2.
8. Drazen JM, Israel E, O'Byrne PM. Treatment of asthma with drugs modifying the leukotriene pathway. N. Engl, J Med 1999; 340:197-206.

9. Knorr B, Franchi LM, Bisgaard H et al. Montelukast, a leukotriene receptor antagonist, for the treatment of persistent asthma in children aged 2 to 5 years. *Pediatrics* 2001;**108**(3):E48.
10. Brodlie M, Gupta A, Rodriguez-Martinez CE et al. Leukotriene receptor antagonists as maintenance and intermittent therapy for episodic viral wheeze in children. Cochrane Database Syst. Rev. Protocol for review: CD008202
11. Brand PL, Baraldi E, Bisgaard H et al. Definition, assessment and treatment of wheezing disorders in pre-school children: an evidence-based approach. Eur Respir J2008;32:1096-110.
12. Ater D, Shad H, BarBE et al. Hypertonic saline and acute wheezing in preschool children. Pediatrics 2012: e1397-40